

NAFIZ SADMAN

sadman.n@queensu.ca | +1 438 867 7717 | [Github](#) | [LinkedIn](#)

RESEARCH EXPERIENCES

Computing Graduate Research Fellow (Big Data Analytics and Management Lab, Queens' School of Computing)
Sept 2022 – Present

■ **Research focus:** Explainable radiology vision-language models for multi-label long-tail cancer identification and visual grounding.

Related publication: Interpreting Biomedical VLMs on High-Imbalance Out-of-Distributions: An Insight into BiomedCLIP on Radiology. In the 24th International Workshop on Datamining in Bioinformatics (BioKDD 25).

■ Cross Affiliation with School of Computing and Faculty of Health Science, Queen's University
Supervisor: Professor Farhana Zulkernine, Ph.D., P.Eng and Dr. Nicholas Cofie, Ph.D.

Automated the extraction and evaluation of student resident performance (Entrustable Professional Activities) by **analyzing textual information** from assessor notes. This includes **sentiment analysis and spearman correlation** of assessor feedback and committee comments.

■ Cross Affiliation with School of Computing and School of Rehabilitation Therapy, Queen's University
Supervisor: Professor Farhana Zulkernine, Ph.D., P.Eng and Assoc Professor Nora Fayed, Ph.D.

1. **Automated literature filtration with domain-specific BERT, information extraction, and information classification from 8,000 clinical RCT articles with specificity 95%** using domain-specific large language model.
2. Contributed to **grant writeup**.

■ **Internship at Pfizer Canada** (with MITACS Accelerate | Oct 2022 - Feb 2023)

1. **Extracted, transformed, and analyzed unstructured Electronic Health Records** to understand patient clinical characteristics, demographics, and comorbidities using **Pandas, MySQL, numpy, sklearn**, and visualization with **seaborn**. Conducted **statistical prediction with k-means clustering and logistic regression**.
2. Contributed to **paper**: "Diagnosis and management of Alzheimer's disease in primary care: a real-world study in Ontario, Canada" accepted in Journal of Primary Care and Community Health (2025).

INDUSTRY EXPERIENCES

Associate Machine Learning Engineer (Internship)

Sep 2025 – Dec 2025

EvenUp Law (US-based AI company for Personal Injury)

- Explored **GEPA** (evolutionary prompt optimization) and its effectiveness.
- **Created a firmwide relevance dataset** for EvenUp with **LLM-as-judge**, that can be used as an evaluation harness and **retrieval metric monitoring (precision@k, recall@k and NDCG@k)**.
- Implemented **cross-encoder, listwise, and setwise reranking** to improve retrieval metrics by **12%** from baseline retrieval.

Front-end Developer (1 year+)

Jan 2024 – December 2025

Smith School of Electrical and Computer Engineering, Queen's University

- Developed a **visualization research tool** for **streaming data** from PID controllers using: **Python Socket IO, HTML5, CSS3, D3 JavaScript**.
- Document tool for **paper publication** on tools (accepted in NFM2025).

Machine Learning Engineer (3 years)

Aug 2019 - Jul 2022

Silicon Orchard Ltd, Dhaka, Bangladesh

- Built an internal **HR data analytics** tool based on **BERT**.
- Developed a **Stylometric Analysis** algorithm for **Authorship Attribution**.
- Developed a **deep Reinforcement Learning** algorithms for **recommendation systems**.
- Developed **speech recognition** algorithm for **smart home systems**

RESEARCH PROJECTS

Computational Biology	Combinatorial Optimization	Reinforcement Learning	Cyber security
Explainable AI in Radiology Imaging using GradCAM. Paper accepted in BioKDD'25.	Multi-task learning to select best group of classifiers for specific set of tasks.	Imbalanced classification for biomedical datasets using RL for weight distribution.	Using multi-party computation to pass secret messages. Biometric authentication.

OTHER EXPERIENCES

Graduate Teaching Assistant

Jan 2023 – Present

School of Computing, Queen's University

Multiple CS courses. Duties: Assist with marking, proctoring, conducting lab sessions, and office hours.

Community Volunteering

COMPSA Liaison Officer, Graduate Computing Society, Queen's University
VP, Queen's Bangladeshi Student Association, Queen's University

January 2025 – August 2025
September 2024 – Present

ACADEMICS

Ph.D. Candidate

Pursuing

School of Computing, Queen's University

XAI in Vision and NLP.

M.Sc in Cybersecurity

Degree Conferred

School of Computing, Queen's University

Thesis: Passive Liveness Detection using Depth and rPPG for Face Presentation Attack Detection

CERTIFICATIONS

- Deep Learning Certification by *Deeplearning.AI*
- TensorFlow in Practice Certification by *Deeplearning.AI*

PUBLICATIONS

Highlights:

- Sadman, N., Alaca, F., Zulkernine, F (2026 January). DepthPulse+: A Depth and Vital Sign Based Method for Face Presentation Attack Detection. IEEE International Conference on Communications (ICC'26) — Track: Communications and Information Systems Security (CISS).
- Sadman, N., Zulkernine, F, Kwan, B (2025 August). Interpreting Biomedical VLMs on High-Imbalance Out-of-Distributions: An Insight into BiomedCLIP on Radiology. In the 24th International Workshop on Datamining in Bioinformatics (BioKDD 25).
- Sadman, N., Kieneris, N., Kauffman, S. (2025 April). Visualizing Temporal Interval Hierarchies. In the 17th NASA Formal Methods (NFM).
- Sadman, N., Hasan, K.A., Rashno, E., Alaca, F., Tian, Y., Zulkernine, F. (2023 December). Vulnerability of Open-source Face Recognition Systems to Blackbox Attacks: A Case Study With InsightFace. In 2023 IEEE Symposium on Computational Intelligence in Cyber Security (IEEE CICS)}.

First Authored Publication: 12 **Venues:** KDD, ICML, ICMLA, SSCI, ICSoft, ICSEA, CCWC

Google Scholar Citations: <https://scholar.google.com/citations?user=M3FEjQsAAAAJ>

REFERENCES

Dr. Farhana Zulkernine, Ph.D., P.Eng.

Professor, School of Computing, Queen's University
farhana.zulkernine@queensu.ca

Dr. Furkan Alaca, Ph.D.

Assistant Professor, School of Computing, Queen's University
furkan.alaca@queensu.ca

Dr. Kishor Datta Gupta, Ph.D.

Assistant Professor, School of Physical Sciences, Clark Atlanta University
CEO, Silicon Orchard Ltd.
kgupta@cau.edu